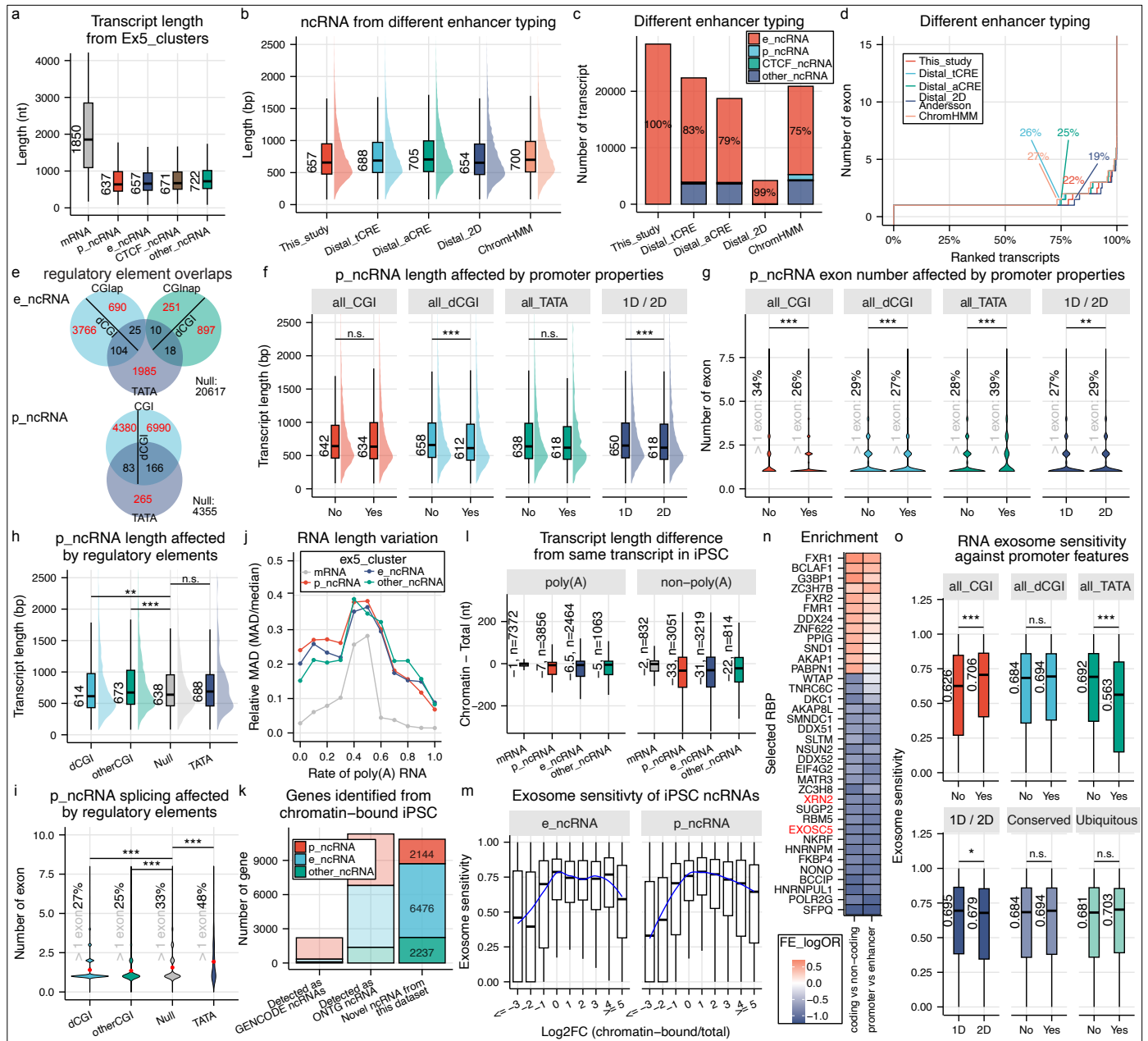




### Extended Data Figure 6 | Factors influencing RNA transcription.

**a**, Transcript lengths of RNA groups defined by ex5\_clusters. **b**, Transcript length of eRNAs defined using various enhancer annotation methods. **c**, Proportion of RNA groups identified in this study among enhancers defined by different methods. **d**, Median exon number of eRNAs by enhancer definition method; percentages indicate the proportion with a median exon count > 1. **e**, Venn diagrams showing overlap of ex5\_clusters across TATA, CGIap, and CGInap, and the subset with downstream CGI (dCGI) for e\_ncRNA. The p\_ncRNA was divided into TATA and CGI with dCGI subset. **f**, Transcript length of p\_ncRNAs compared across promoter features (as defined in **Figure 4c**). **g**, Exon count of p\_ncRNAs across promoter features; percentages indicate groups with more than one exon. **h**, Transcript length of four mutually exclusive p\_ncRNA groups, based on individual promoter features. **i**, Median exon number for the four mutually exclusive p\_ncRNA groups; percentages indicate groups with more than one exon. **j**, Length variation within ex5\_clusters represented by relative MAD (MAD/median); plotted alongside poly(A) RNA rate (defined in **Figure 4k**). **k**, SALA analysis of chromatin-bound CFC-seq from iPSCs using the finalized transcriptome plus GENCODE v39. Shown are ncRNA genes originating from GENCODE, the novel ncRNA identified from the finalized transcriptome (ONTG), or the chromatin-bound library alone. **l**, Transcript length differences for shared transcript models between the chromatin-bound and main datasets, grouped by poly(A) status in the main dataset. **m**, Exome sensitivity of transcript models (1,935 e\_ncRNAs and 3,841 p\_ncRNAs) with CPM > 0.25, based on RNA-seq of EXOSC3 knockdown in iPSC. Sensitivity ( $\log_2$  fold change) was estimated using edgeR on chromatin-bound vs. total RNA CFC-seq (quantified by bambu with the SALA finalized transcriptome updated with chromatin-bound models). **n**, Enrichment of individual RBP interactions from exons of coding vs. non-coding, and enhancer- vs. promoter-derived transcripts. **o**, Enrichment of exome sensitivity in p\_ncRNAs across different promoter features (defined in **Figure 4n**). Unless specified, statistical tests were two-sided Wilcoxon;  $p < 0.05$  (\*),  $p < 0.01$  (\*\*),  $p < 0.001$  (\*\*\*), n.s. = not significant.